

Vibe Atlas

System Design Document — Architecture, Scoring, Search & UX Specification

Document Type: Internal Technical Reference — Confidential

Version: 1.5 | **Status:** Active — Revised July 15, 2026

Date: July 9, 2026

Audience: Senior Engineers, Product Managers, Technical Leads

Prepared by: Platform Architecture Team

Table of Contents

1. Executive Summary

1.1 Platform Mission | 1.2 Core Problem | 1.3 Architectural Bet | 1.4 Design Principles

2. System Architecture Overview

2.1 High-Level Component Diagram | 2.2 Data Flow Narrative | 2.3 Technology Stack
| 2.4 Scalability Targets

3. Grid Classification System

3.1 Grid Classes | 3.2 Classification Algorithm | 3.3 Grid Packing | 3.4 Viewport
Breakpoints | 3.5 Editorial Pin System | 3.6 Re-Classification Triggers

4. Quality Score Engine

4.1 Composite Quality Score | 4.2 Sub-Score Definitions | 4.3 Update Cadence | 4.4
Score Bands & Eligibility | 4.5 Penalties | 4.6 Data Pipeline

5. Quality Heuristics

5.1 Rationale | 5.2 Heuristic Categories | 5.3 Application Rules | 5.4 Quality Council

6. Promo Detection

6.1 Threat Model | 6.2 Signal Groups | 6.3 Ensemble Classifier | 6.4 Decision
Thresholds | 6.5 Human Review Workflow | 6.6 Creator Appeals | 6.7 Model
Governance

7. Search Strategy

7.1 Search Modes | 7.2 Query Understanding Service | 7.3 Retrieval | 7.4 Ranking | 7.5 Result Shaping | 7.6 Vibe Taxonomy | 7.7 Autocomplete | 7.8 Telemetry

8. Height Emphasis System

8.1 Purpose | 8.2 Base Height Formula | 8.3 HeightEmphasisFactor Inputs | 8.4 Aspect Ratio Policies | 8.5 Animation & Transition Rules | 8.6 Grid Reflow Strategy

9. Connect Hub Integration

9.1 What is Connect Hub | 9.2 Bidirectional Bridge | 9.3 Data Flows & Payload Schemas | 9.4 Authentication | 9.5 SLA Commitments

10. UX Specification

10.1 Design Principles | 10.2 Grid Card Anatomy | 10.3 Card States | 10.4 Search UX | 10.5 Vibe Filter Panel | 10.6 Content Detail Page | 10.7 Connect Hub Creator Surfaces | 10.8 Accessibility

11. Implementation Phases

Phase 0-4 Breakdown | Phase Gate Governance | Cross-Phase Dependencies

12. Open Questions and Future Work

1. Executive Summary

1.1 Platform Mission

Vibe Atlas is a spatially organized, quality-driven content discovery platform for music and cultural content. Its defining characteristic is a dynamic, living grid surface in which the size, position, and visual weight of each content card is a direct function of computed quality — not of paid placement, recency alone, or raw popularity. The grid is the product: it communicates editorial judgment at a glance, rewards authentic creativity, and provides users with a discovery experience that feels curated without being opaque.

The platform serves three constituencies simultaneously. Listeners and fans receive a discovery surface that surfaces genuinely excellent work, not merely trending or promoted content. Artists and creators receive transparency into how their work is ranked and tools to understand — and legitimately improve — their standing. Curators and editorial teams receive a first-class set of override and annotation

tools that let human judgment complement algorithmic classification without overriding its integrity.

1.2 The Core Problem

The dominant content discovery paradigm across music streaming, social video, and cultural media platforms has converged on engagement optimization: maximize clicks, plays, shares, and time-on-platform as proxied signals for content quality. This paradigm contains a structural flaw: engagement metrics are gameable. Velocity-injected plays, bot-driven saves, coordinated cross-platform promotion, and paid follower networks all produce engagement signals that are indistinguishable from organic discovery at the raw data layer. Over time, platforms optimizing for these signals amplify inauthentic content, train their recommendation models on corrupted data, and erode user trust in the discovery experience.

The secondary problem is opacity. Creators whose content is demoted or suppressed have no mechanism to understand why, no recourse to contest erroneous classification, and no signal about what authentic improvement looks like. This asymmetry creates antagonism between creators and platforms that serves neither party.

1.3 The Architectural Bet

Vibe Atlas makes three architectural commitments that differentiate it from engagement-first platforms:

1. **Compute quality holistically at index time.** Rather than ranking content reactively at query time using raw engagement signals, Vibe Atlas computes a composite Quality Score for every piece of content on a continuous basis. The score integrates engagement, retention, authenticity, and freshness — four dimensions that are harder to game in concert than any single metric. The score is the first-class indexing artifact; raw engagement data is an input, not the output.
2. **Make the grid a living surface driven by real-time signals.** The discovery grid is not a static ranked list. Card dimensions, visual emphasis, and grid position update as quality scores change. A track experiencing a

genuine resurgence physically grows on the grid. A content item flagged for inauthentic promotion shrinks or disappears. The grid surface is the visual expression of the scoring system, not a layer applied on top of it.

3. **Give creators transparency into how their content is ranked.** Through the Connect Hub integration, creators receive structured score explanations, sub-score breakdowns, flag notifications with evidence, and a formal appeal pathway. Transparency is a first-class product feature, not an afterthought.

1.4 Design Principles

Four complementary forces govern every significant design decision in this document:

Principle	What It Means in Practice	Where It Appears
Authenticity	Organic signals are weighted more than promotional ones. Inauthentic behavior is detected, penalized, and disclosed.	Promo Detection, Authenticity sub-score, Penalty table
Momentum	Content that is gaining genuine traction is rewarded with greater visual prominence, regardless of absolute score level.	Freshness Score, Resurgence Boost, Momentum Factor in Height Emphasis
Personalization	Discovery is shaped by individual listening context, not only global quality rankings.	Search personalization feature group, Personalization Factor in Height Emphasis, collaborative filtering
Editorial Curation	Human judgment is a legitimate, bounded input to the grid. Editors can amplify content within defined constraints but cannot arbitrarily suppress organic rankings.	Editorial pin system, Editorial Emphasis Score, time-boxed overrides

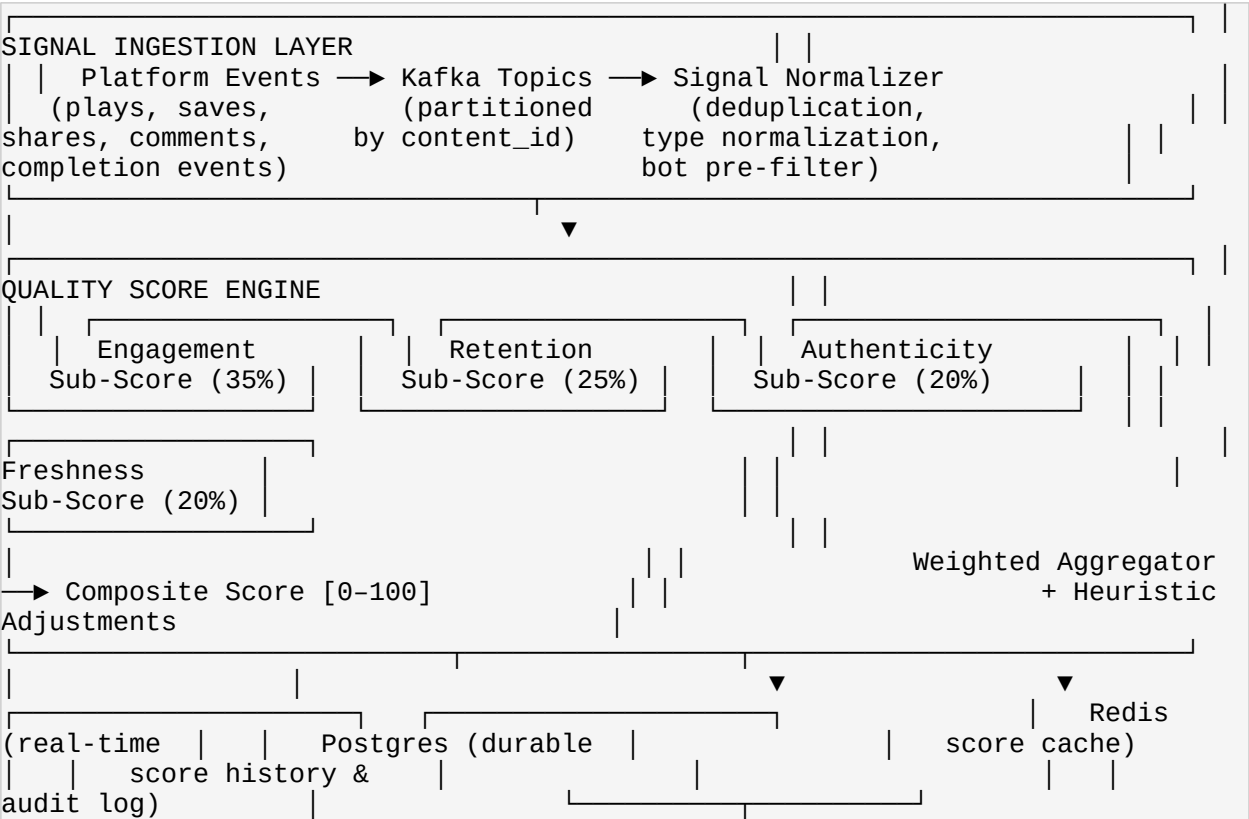
Document Scope

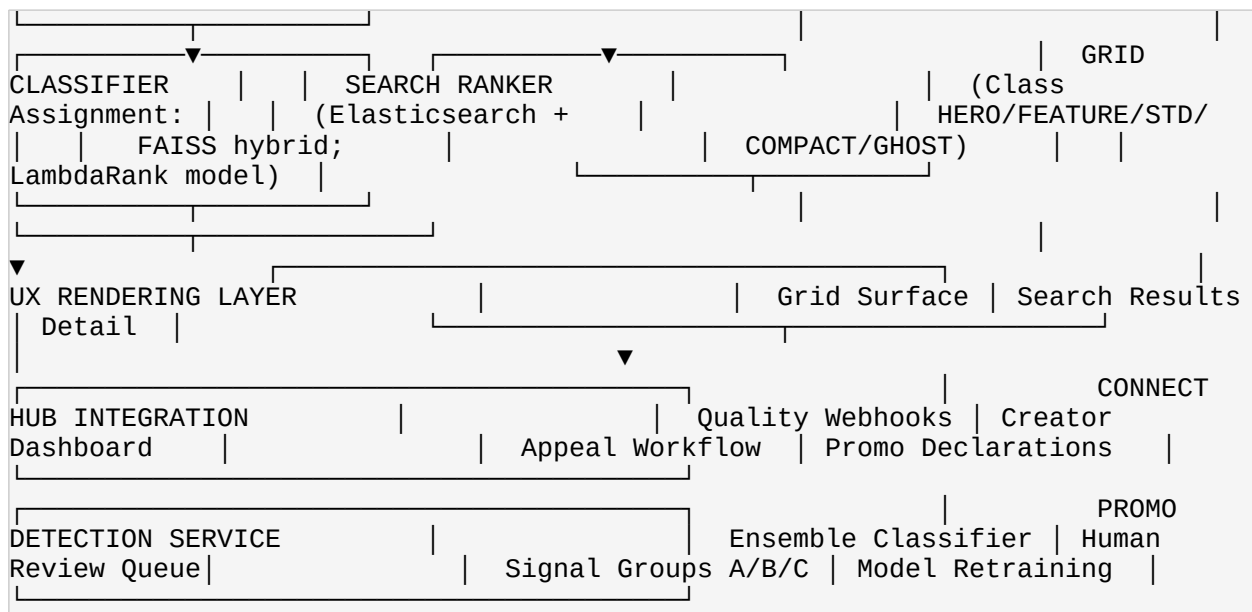
This document covers the full architectural surface of Vibe Atlas: ingestion, scoring, classification, search, UX, creator tooling, and implementation phasing. It is intended as a living reference for engineering, product, and technical leadership. Sub-components may produce their own detailed technical specifications; this document governs the interfaces and contracts between them.

2. System Architecture Overview

2.1 High-Level Component Diagram

The following ASCII diagram represents the top-level data flow through the Vibe Atlas platform. Each box corresponds to a major subsystem; arrows represent primary data flow directions. Bidirectional arrows indicate systems with both inbound and outbound integration responsibilities.





2.2 Data Flow Narrative

The lifecycle of a piece of content — from initial publish to appearance in the discovery grid — follows a deterministic pipeline with defined latency budgets at each stage.

Step 1 — Ingestion: When a user action (play, save, share, comment) or a platform event (content publish, DMCA notice) occurs, the event is published to the appropriate Kafka topic. Topics are partitioned by `content_id` to ensure all signals for a given item are consumed by the same downstream consumer group, preserving ordering guarantees needed for deduplication. The Signal Normalizer consumes events, applies deduplication based on event fingerprints (user × content × event_type × 5-minute window), normalizes signal types to a canonical schema, and runs a lightweight bot pre-filter (IP velocity, user-agent heuristics) before forwarding to sub-score calculators.

Step 2 — Scoring: Sub-score calculators maintain running aggregates per content item in Redis. On each signal arrival, the relevant sub-score (Engagement, Retention) is recomputed and the composite Quality Score is updated. The full re-score including Authenticity and Freshness components runs on a 6-hour batch cycle. Scores are written to both Redis (for low-latency reads by the Grid Classifier) and Postgres (for durable history, audit logs, and the Connect Hub webhook trigger).

Step 3 — Classification: The Grid Classifier reads the updated composite score from Redis, applies the classification algorithm (§3.2), and determines the content

item's grid class. If the class has changed from the previous assignment, a re-classification event is published, triggering a grid repack for any grid instance that includes the item. The classifier also notifies the Search Ranker to reindex the item's quality feature vector.

Step 4 — Serving: The UX Rendering Layer assembles grid pages from pre-computed grid layouts stored in Redis, applying real-time Height Emphasis factors (§8) and personalization overlays before response serialization. Search results are assembled by the Search Ranker at query time using the four-stage pipeline described in §7.

Step 5 — Connect Hub Notification: When a score update exceeds the delta threshold (configurable per creator tier), or when a promo flag is raised, the Connect Hub bridge emits a webhook to the creator's registered endpoint and updates the creator dashboard data store. Appeals and metadata updates flow in the opposite direction: Connect Hub → Atlas ingestion pipeline.

2.3 Technology Stack

Component	Technology Choice	Rationale
Signal streaming	Apache Kafka	High-throughput, durable, partitioned event streaming with strong ordering guarantees per partition. Proven at scale for event-driven scoring pipelines.
Lexical search	Elasticsearch (BM25+)	Mature, horizontally scalable full-text retrieval with BM25+ scoring, field boosting, and faceted filtering. Native support for index aliases enables zero-downtime reindexing.
Semantic search	FAISS (768-dim BERT embeddings)	Sub-millisecond approximate nearest-neighbor search at billion-vector scale. FAISS IVF-PQ indices fit within GPU memory for low-latency embedding lookups.
Score cache	Redis (sorted sets)	Sub-millisecond reads for Grid Classifier and UX Rendering Layer. Sorted

Component	Technology Choice	Rationale
		sets support efficient range queries for score band filtering.
Score persistence	PostgreSQL	ACID-compliant audit log and score history. Serves as the source of truth for Connect Hub webhook triggers and appeal evidence packages.
ML services	Python (PyTorch, scikit-learn, Hugging Face)	Mature ML ecosystem for LambdaRank, ensemble classifiers, BERT fine-tuning, and collaborative filtering. Python services deployed as containerized gRPC microservices.
Grid state management	Redis (hash maps + pub/sub)	Grid layout state is stored as Redis hashes per grid instance. Pub/sub channels notify UX rendering workers of re-classification events requiring grid repack.
API layer	Python (FastAPI)	High-performance async API framework with automatic OpenAPI documentation. Native Pydantic validation for Connect Hub payload schemas.
Orchestration	Kubernetes + Helm	Container orchestration with horizontal pod autoscaling keyed to Kafka consumer lag metrics for the scoring pipeline.
Observability	Prometheus + Grafana + OpenTelemetry	Unified metrics, distributed tracing, and alerting across all pipeline stages. Latency SLOs enforced via Prometheus alerting rules.

2.4 Scalability Targets

Performance Envelope

All system components must be designed to sustain the following targets under peak

load. These are hard SLOs, not aspirational goals. Each phase gate (§11) includes load test validation against these figures before promotion to production.

- **Concurrent users:** 50,000 sustained; 75,000 burst (1.5× multiplier)
- **API latency:** $P99 \leq 300\text{ms}$ for grid page assembly; $P99 \leq 150\text{ms}$ for search result retrieval
- **Score update throughput:** $\geq 50,000$ signal events per second across all Kafka topics
- **Grid re-classification SLA:** $P95 \leq 90$ seconds from score delta to updated grid class in Redis
- **Webhook delivery SLA:** $P95 \leq 5$ seconds from score delta trigger to webhook HTTP delivery
- **Autocomplete latency:** $P99 \leq 80\text{ms}$ end-to-end
- **Availability target:** 99.9% monthly uptime (excluding planned maintenance windows)

3. Grid Classification System

The grid classification system is the mechanism by which every piece of content on Vibe Atlas is assigned a visual weight — a grid class — that determines how much space it occupies on the discovery surface. Classification is not merely a UI decision; it is the primary feedback mechanism by which the quality scoring system communicates its judgments to users. The grid is the score made visible.

3.1 The Five Grid Classes

Class	Size Multiplier	Score Requirement	Additional Trigger Conditions	Eligibility Constraints
HERO	4 × 4 grid units	Quality Score ≥ 85 (Platinum band)	Editorial pin OR Momentum Factor ≥ 0.90 OR trending rank ≤ 5 in category	≤ 2 HERO cards per grid page; no two adjacent HERO cards; requires Authenticity sub-score ≥ 70
FEATURE	2 × 2 grid units	Quality Score 70–84 (Gold band)	Score 65–69 with Momentum Factor ≥ 0.80 OR editorial soft-pin	≤ 6 FEATURE cards per grid page; Authenticity sub-score ≥ 55
STANDARD	1 × 1 grid units	Quality Score 45–69 (Silver/Bronze band)	Default class for eligible, non-suppressed content	No additional constraints; majority of grid population
COMPACT	1 × 0.5 grid units	Quality Score 20–44 (Bronze band, lower range)	Cold-start content with insufficient signal; heuristic-adjusted items in lower bands	No column may contain more than 3 consecutive COMPACT cards
GHOST	Hidden (0 × 0)	Quality Score < 20 (Suppressed band) OR Promo Detection score ≥ 0.85	Active DMCA hold; community report threshold exceeded; human review pending	Content remains indexed but is excluded from all organic discovery surfaces; appears only in direct search by exact title

Design Decision: GHOST vs. Removal

GHOST class deliberately preserves content in the index rather than deleting it. This is intentional: removal is irreversible and removes the evidence trail needed for appeals. GHOST items remain discoverable via direct title search so that creators can verify the item still exists and initiate an appeal via Connect Hub. Removal from the index requires a separate explicit action (e.g., DMCA takedown confirmed by legal review) and is outside the scope of the classification system.

3.2 The Classification Algorithm

The classifier runs as an async worker consuming from the `score-updates` Kafka topic. Every score update event triggers evaluation of the classification function. The algorithm is deterministic given the same inputs; all randomness is excluded to ensure auditability.

```
function classify_content(content_item):    score    =
content_item.quality_score                # float [0, 100]    auth_score =
content_item.authenticity_sub_score       # float [0, 100]    momentum  =
content_item.momentum_factor              # float [0, 1]      flags      =
content_item.active_flags                  # set of flag types  editorial  =
content_item.editorial_override            # None | OverrideRecord trending_rank
= content_item.trending_rank                # int | None      # --- Step 1:
Suppression checks (highest priority) ---    if "DMCA_HOLD" in flags:
return GHOST, reason="dmca_hold"              if "PROMO_CONFIRMED" in flags and
content_item.promo_score >= 0.85:              return GHOST,
reason="promo_confirmed"                      if "COMMUNITY_REPORT_THRESHOLD" in flags:
return GHOST, reason="community_reports"      if score < 20:          return
GHOST, reason="low_quality_score"              # --- Step 2: Editorial override check
(time-boxed) ---    if editorial is not None and editorial.is_valid():
target_class = editorial.target_class          # Heuristics cannot push to
HERO via editorial; HERO requires              # organic score >= 85 OR explicit
HERO editorial pin with approval              if target_class == HERO and score <
85:          if not editorial.has_hero_approval:
target_class = FEATURE # cap to FEATURE          return target_class,
reason="editorial_override"                    # --- Step 3: HERO eligibility ---    if
score >= 85 and auth_score >= 70:              hero_trigger =
(          (momentum >= 0.90) or              (trending_rank is not None
and trending_rank <= data-id="228" 5)          )          if hero_trigger:
return HERO, reason="organic_hero"              # --- Step 4: FEATURE eligibility ---
if score >= 70 and auth_score >= 55:          return FEATURE,
reason="organic_feature"                      if score >= 65 and auth_score >= 55 and
momentum >= 0.80:          return FEATURE, reason="momentum_feature"          #
--- Step 5: STANDARD eligibility ---    if score >= 45:          return
STANDARD, reason="organic_standard"            # --- Step 6: COMPACT eligibility
---    if score >= 20:          return COMPACT, reason="organic_compact"
# --- Fallback (should not reach here after Step 1 check) ---    return
GHOST, reason="fallback_suppression"
```

3.3 The Grid Packing Algorithm

Once content items have been classified, the grid packer assembles them into a two-dimensional grid layout subject to anti-clustering constraints. The algorithm is a modified bin-packing approach that treats the grid as a column-major matrix of unit cells.

Anti-Clustering Rules

- **No two adjacent HERO cards:** HERO cards (4×4) must be separated by at least one column of non-HERO content. The packer enforces this by marking a 1-column exclusion zone to the right of each placed HERO card.
- **No column with more than 3 consecutive COMPACT cards:** The packer tracks per-column consecutive COMPACT counts. When the count would reach 4, a STANDARD or higher card is injected from the pending queue.
- **FEATURE card distribution:** No more than 2 FEATURE cards may appear in adjacent columns at the same vertical position, preventing the grid from feeling visually monotonous at mid-tier density.
- **GHOST cards:** Not placed in the grid. The packer's input queue filters GHOST items before layout computation begins.

```
function pack_grid(classified_items, viewport_columns):    grid =
Grid(columns=viewport_columns)    hero_count = 0    pending =
sort_by_quality_score(classified_items)    for item in pending:        if
item.class == HERO:            if hero_count >= 2:
item.class = FEATURE # demote if HERO quota exceeded            else:
position = grid.find_hero_slot(exclusion_zones)            if position
is None:                item.class = FEATURE # no valid slot; demote
else:                grid.place(item, position, size=(4,4))
grid.mark_exclusion(position, width=1)                hero_count += 1
continue            if item.class == COMPACT:                col =
grid.next_available_column()                if
grid.consecutive_compact_count(col) >= 3:                    # Force a STANDARD
injection before placing COMPACT                    standard_item =
pending.pop_next_eligible(min_class=STANDARD)                    if
standard_item:                        grid.place(standard_item,
grid.next_slot(col), size=(1,1))                        grid.place(item,
grid.next_slot(col), size=(1,0.5))                        continue
grid.place(item, grid.next_best_slot(), size=item.class.size)                return
grid.layout
```

3.4 Viewport Breakpoints and Responsive Behavior

Breakpoint	Label	Width Range	Grid Columns	HERO Max	Notes
xs	Mobile Portrait	0 - 479px	2	0 (hidden)	HERO class rendered as FEATURE; COMPACT renders at 1×1

Breakpoint	Label	Width Range	Grid Columns	HERO Max	Notes
sm	Mobile Landscape	480 – 767px	4	1	HERO rendered at 2×2 (reduced); anti-clustering rules relaxed
md	Tablet	768 – 1023px	6	1	Full HERO size at 3×3 equivalent; standard packing rules apply
lg	Desktop	1024 – 1439px	8	2	Full packing algorithm; all anti-clustering rules active
xl	Wide Desktop	≥ 1440px	12	2	Extended gutter; Height Emphasis factor scales up to 1.25× maximum

3.5 Editorial Pin and Override System

Curators with appropriate role permissions may create time-boxed editorial overrides that assign a specific grid class to a content item, bypassing the organic classification algorithm. Overrides are not permanent: they carry an expiry timestamp, and on expiry the item reverts to organic classification without manual intervention.

Key constraints on editorial overrides:

- **Time-boxing:** All overrides must specify a duration (minimum 1 hour, maximum 30 days). Overrides with no expiry are rejected by the system.
- **HERO approval gate:** Elevating content to HERO class via editorial pin requires secondary approval from a Senior Curator. The system enforces this via an approval workflow in the Admin Console; un-approved HERO pins default to FEATURE.

- **Suppression protection:** Editorial overrides cannot lift content out of GHOST class if the GHOST classification was triggered by an active DMCA hold or confirmed promo violation. Legal review must clear the flag before editorial tools can act.
- **Audit log:** Every override creation, modification, and expiry is written to an immutable audit log in Postgres with curator identity, timestamp, and justification note.
- **Sponsored labeling:** Any editorial pin associated with a commercial arrangement (paid placement) must include a `sponsored=true` flag. This flag triggers the "Sponsored" badge in the UX and is enforced at the API level — it cannot be omitted by client applications.

3.6 Re-Classification Triggers and Async Worker SLA

Re-classification is triggered by any of the following events:

- Composite Quality Score delta exceeds ± 5 points from the score recorded at last classification
- Promo Detection score exceeds any threshold boundary (see §6.4)
- Editorial override created, modified, or expired
- Active flag status change (DMCA hold applied or lifted; community report threshold crossed)
- Forced re-classification requested via Admin Console

The async worker pool processing re-classification events must sustain a **P95 end-to-end latency of ≤ 90 seconds** from trigger event publication on the Kafka topic to updated class value written to Redis. This SLA is monitored by a Prometheus alerting rule; breach of $P99 \leq 3$ minutes triggers an on-call page.

4. Quality Score Engine

The Quality Score Engine is the computational heart of Vibe Atlas. It transforms raw engagement, behavioral, and contextual signals into a single composite score — a float in the range [0, 100] — that serves as the primary ranking artifact across the grid classification system, search ranker, and Connect Hub transparency tools. The engine is designed to be difficult to game: because the score integrates four conceptually distinct dimensions, optimizing for one at the expense of others is structurally visible as a signal anomaly.

4.1 Composite Quality Score Formula

The composite Quality Score **Q** is a weighted linear combination of four sub-scores, each independently normalized to [0, 100]:

$$Q = (0.35 \times E) + (0.25 \times R) + (0.20 \times A) + (0.20 \times F)$$

Where: E = Engagement Score [0, 100] – weight: 35% R = Retention Score [0, 100] – weight: 25% A = Authenticity Score [0, 100] – weight: 20% F = Freshness Score [0, 100] – weight: 20%

Heuristic adjustments (§5) are additive post-aggregation, subject to band ceiling protection and cap constraints defined in §5.3. The raw Q value before heuristic adjustment is stored separately to allow full auditability of the adjustment contribution.

4.2 Sub-Score Definitions

4.2.1 Engagement Score (E) — Weight: 35%

The Engagement Score measures the breadth and depth of user interaction with a content item, normalized against peer content in the same category and release cohort to control for inherent popularity differences between genres and audience sizes.

Signal	Raw Weight	Normalization	Rationale
Saves / bookmarks	3.0	Peer-normalized (category cohort)	High-intent signal; requires deliberate action; strongly predictive of re-engagement
Shares (external +	2.5	Peer-normalized	Social amplification;

Signal	Raw Weight	Normalization	Rationale
internal)			external shares weighted 1.3x internal shares
Comments	2.0	Peer-normalized; sentiment-filtered	High intent; negative-sentiment comments excluded from positive signal but tracked separately for community moderation
Plays (qualified)	1.5	Peer-normalized; bot-filtered	Qualified play = play lasting $\geq 20\%$ of track duration; raw play counts are not used directly
Likes / reactions	1.0	Peer-normalized	Low-friction; weighted lowest due to susceptibility to automated inflation

Peer normalization uses a z-score transformation against a rolling 30-day distribution of the same signals for content in the same category and release cohort (content published within the same 90-day window). This ensures that a track receiving 1,000 saves in a niche genre competes fairly against a track receiving 100,000 saves in a mainstream genre.

4.2.2 Retention Score (R) — Weight: 25%

The Retention Score measures whether users who encounter content are compelled to stay with it and return to it. It is composed of three components:

- **Completion Rate (50% of R):** The median percentage of track/content duration consumed per qualified play session. Bucketed into deciles and scored 0–100. A completion rate at the 90th percentile of the peer cohort scores 90+.
- **Re-engagement Rate (30% of R):** The proportion of users who play or save the same content item more than once within a 30-day window. High re-engagement is a strong proxy for genuine quality independent of initial discovery context.

- **Session Depth (20% of R):** The average number of additional content items consumed by users in the same session after encountering this item. Content that deepens session engagement — rather than ending it — receives a positive signal.

4.2.3 Authenticity Score (A) — Weight: 20%

The Authenticity Score quantifies the probability that a content item's engagement signals are organic. It is the primary defense against score inflation via inauthentic promotion. A low Authenticity Score is a strong suppressive force on the composite Q, regardless of raw engagement volume.

- **Bot Probability (inverse, 40% of A):** $A_{\text{bot}} = 1 - P(\text{bot} \mid \text{engagement_pattern})$, derived from the Promo Detection ensemble classifier (§6). High bot probability directly reduces A.
- **Source Diversity Index (30% of A):** Herfindahl-Hirschman Index (HHI) applied to the geographic, platform-referral, and user-demographic distribution of signal sources. Organic discovery produces diverse, geographically distributed signals; coordinated promotion produces concentrated ones.
- **Velocity Anomaly Score (inverse, 20% of A):** Flags sudden, unsustained engagement spikes inconsistent with the item's historical trajectory. A content item accumulating its entire 30-day engagement in a 6-hour window receives a high velocity anomaly score, which suppresses A.
- **Promo Detection Confidence (inverse, 10% of A):** Direct feed from the Promo Detection service output probability. Values above 0.50 linearly suppress A; values above 0.85 zero out A entirely (triggering GHOST classification).

4.2.4 Freshness Score (F) — Weight: 20%

The Freshness Score rewards recency while penalizing staleness, using category-tuned exponential decay to prevent older high-quality content from monopolizing the grid indefinitely.

$F_{base} = 100 \times e^{(-\lambda \times \text{age_in_days})}$ Category-specific decay constants (λ):
Viral / trending content: $\lambda = 0.15$ (half-life ≈ 4.6 days) New releases (pop/hip-hop): $\lambda = 0.07$ (half-life ≈ 9.9 days) Album-cycle content: $\lambda = 0.03$ (half-life ≈ 23 days) Evergreen / catalog: $\lambda = 0.01$ (half-life ≈ 69 days) Archived / legacy: $\lambda = 0.005$ (half-life ≈ 139 days)

Resurgence Boost Mechanism: When a content item's engagement velocity in the trailing 7 days exceeds $2\times$ its trailing 30-day baseline, the Freshness Score receives a resurgence boost: $F = \min(F_{base} \times 1.40, 100)$. This mechanism ensures that genuinely rediscovered catalog content can re-enter competitive grid positions without requiring a new release.

4.3 Score Update Cadence

Update Type	Trigger	Cadence / Latency	Components Updated
Real-time update	Signal event arrives on Kafka topic	< 60 seconds end-to-end	Engagement Score, Retention Score (completion events)
Batch update	Scheduled timer	Every 15 minutes	All four sub-scores; composite Q recomputed; Freshness decay applied
Full re-score	Scheduled timer	Every 6 hours	Full pipeline including Authenticity Score (requires Promo Detection inference), heuristic re-evaluation, Postgres audit record
Forced re-score	Admin Console or flag status change	On-demand; queued behind real-time updates	All components; triggers immediate re-classification

4.4 Score Bands and Grid Class Eligibility

Band	Score Range	Grid Classes Eligible	Connect Hub Label
Platinum	85–100	HERO, FEATURE, STANDARD	Platinum

Band	Score Range	Grid Classes Eligible	Connect Hub Label
Gold	70–84	FEATURE, STANDARD	Gold
Silver	55–69	STANDARD	Silver
Bronze	20–54	STANDARD, COMPACT	Bronze
Suppressed	0–19	GHOST only	Suppressed

4.5 Penalties Table

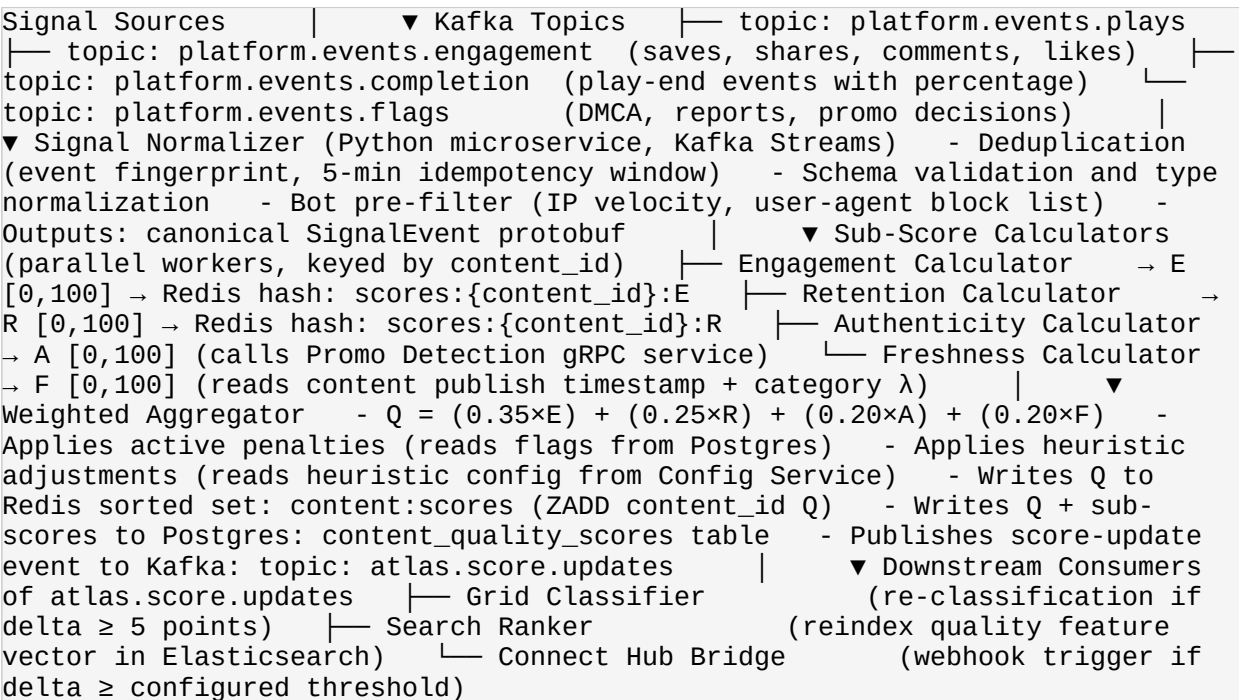
Penalties are applied as subtractive adjustments to the composite Q score before band assignment. Penalty adjustments do not affect sub-score values stored in the audit log — the unadjusted sub-scores are always preserved for transparency and appeal purposes.

Penalty Type	Trigger Condition	Q Score Adjustment	Duration	Override Path
DMCA Notice	Valid DMCA takedown notice received	GHOST (immediate)	Until legal review clears	Legal team via Admin Console
Community Reports (Level 1)	≥ 25 unique user reports in 24 hours	–15	72 hours or until review	Moderation review
Community Reports (Level 2)	≥ 100 unique user reports in 24 hours	GHOST + human review queue	Until review decision	Senior moderation review
Promo Detection (Medium)	Promo score 0.60–0.84	–20 to –35 (scaled)	Until next full re-score clears	Creator appeal via Connect Hub
Promo Detection (High)	Promo score ≥ 0.85	GHOST	Until human review decision	Human reviewer + creator appeal
Promo History Strike	Confirmed promo violation in trailing 12 months	–10 (ongoing, applied to all content by creator)	12 months from confirmation date	Expires automatically; cannot be appealed
Metadata Inconsistency	Submitted metadata fails validation (genre mismatch,	–5	Until metadata corrected via Connect Hub	Creator metadata update

Penalty Type	Trigger Condition	Q Score Adjustment	Duration	Override Path
	invalid ISRC, duplicate title attack)			

4.6 Data Pipeline

The Quality Score Engine data pipeline is the operational backbone connecting raw signal events to scored, indexed content. The pipeline must sustain high throughput with low latency while providing durable state for audit and appeal purposes.



5. Quality Heuristics

5.1 Why Heuristics Exist Alongside the Statistical Score

The composite Quality Score is a powerful and auditable ranking signal, but it has two structural blind spots that heuristics are designed to address.

The Cold Start Problem: Newly published content has no engagement history. A purely data-driven score for a brand-new release from a verified, established artist would be identical to a score for a brand-new release from an unknown creator with no prior activity — both would receive near-zero scores and land in COMPACT or GHOST class regardless of intrinsic quality. Heuristics allow the system to make reasonable prior assumptions based on creator reputation, content metadata, and external signals that exist before organic data accumulates.

Edge Cases the Data Cannot See: Real-world context matters in ways that engagement signals cannot capture automatically. A track nominated for a Grammy Award, a song that scores on a Billboard chart, or a creator who has been the victim of a coordinated harassment campaign leading to artificially elevated community reports — these are factual contexts that the statistical model cannot infer from engagement data alone. Heuristics provide a structured, auditable mechanism for injecting these real-world facts into the ranking system without circumventing the scoring architecture.

Design Decision: Heuristics as Bounded Adjustments

Heuristics are deliberately constrained: they cannot override band boundaries for suppression decisions, cannot elevate content to HERO class on their own, and are subject to a total bonus cap and penalty cap. This prevents heuristic abuse — whether accidental or adversarial — from producing outcomes inconsistent with the statistical quality signal. A heuristic that would push content into HERO territory is capped at FEATURE; a heuristic penalty cannot exceed the Suppressed band floor unless it is itself a flag-based suppression (which is handled by the penalty system, not the heuristic system).

5.2 Heuristic Categories and Key Rules

Category 1: Cold Start Heuristics

Heuristic Rule	Condition	Adjustment	Expiry
Verified artist new release	Creator has verified badge; content	+20	14 days from publish, or until

Heuristic Rule	Condition	Adjustment	Expiry
	published within 7 days; creator has ≥ 1 prior content item with Gold+ score		score reaches Silver band organically
First-time creator	Creator has no prior content items on platform; content passes metadata validation	+5	30 days from publish
Label-submitted content	Content submitted via verified distributor / label Connect Hub account	+8	21 days from publish
Cold start data threshold	Content has fewer than 100 qualified plays	+10 (baseline floor)	Until 100 qualified plays threshold crossed

Category 2: Content Quality Heuristics

Heuristic Rule	Condition	Adjustment	Expiry
High audio fidelity	Content meets lossless or HiFi audio quality standard (verified via metadata + audio analysis)	+5	Permanent while content remains at that quality tier
Rich metadata completeness	Content has complete ISRC, genre tags, mood tags, credits, and artwork above minimum resolution	+3	Permanent; removed if metadata degrades
Duplicate content penalty	Content flagged as near-duplicate (audio fingerprint match ≥ 0.92) of existing indexed content not by same creator	-25	Until fingerprint dispute resolved via Creator appeal

Category 3: Community and Social Heuristics

Heuristic Rule	Condition	Adjustment	Expiry
Editorial playlist inclusion	Content included in an official Vibe Atlas editorial playlist	+12	Duration of playlist inclusion
User-created playlist density	Content appears in ≥ 500 user-created playlists	+8	Recomputed on 15-minute cadence
Harassment-driven report spike	Moderation review determines reports resulted from coordinated harassment, not legitimate policy violations	+10 (restoration credit)	90 days

Category 4: Creator Reputation Heuristics

Heuristic Rule	Condition	Adjustment	Expiry
Promo violation history	Creator has ≥ 1 confirmed promo violation in trailing 12 months	-10 (applied to all content by creator)	12 months from confirmation date
Verified independent artist	Creator is verified as independent (no label affiliation) with consistent release history (≥ 3 releases)	+5	Permanent while status holds
Creator good standing streak	Creator has no violations and ≥ 2 Gold+ content items in trailing 6 months	+5	Rolling 6-month window

Category 5: Real-World Events Heuristics

Heuristic Rule	Condition	Adjustment	Expiry
Award recognition	Content or creator wins or is nominated for a recognized industry award (Grammy, Mercury Prize, etc.)	+15 (win); +8 (nomination)	365 days from award date
Chart recognition	Content appears on	+15 per chart (max	Duration of chart

Heuristic Rule	Condition	Adjustment	Expiry
	a recognized chart (Billboard Hot 100, Official UK Charts, etc.)	3 charts counted)	presence + 14 days
Cultural moment sync	Content associated with a verified cultural event (film/TV sync, major sporting event, etc.), confirmed via metadata tag and editorial approval	+10	90 days from event date
Press coverage spike	Content or creator is subject of significant press coverage (detected via media monitoring feed integration; editorial confirmation required)	+5	30 days from coverage date

5.3 Application Rules

- Total bonus cap: +30.** The sum of all positive heuristic adjustments applied to a single content item at any given time is capped at +30 points. Adjustments beyond this cap are accumulated in a queue and applied as prior adjustments expire, up to the cap.
- Total penalty cap: –50.** The sum of all negative heuristic adjustments (excluding flag-based GHOST penalties) is capped at –50 points. Content cannot be pushed below $Q = 0$ via heuristics alone.
- Band ceiling protection:** Heuristic adjustments cannot push content into the HERO band ($Q \geq 85$). The effective maximum for heuristic-adjusted content is $Q = 84$ (top of Gold band). HERO status requires organic scoring at the Platinum level.
- Expiry and recomputation:** All heuristic adjustments carry explicit expiry timestamps. An expiry worker scans active adjustments every 5 minutes and removes expired rules; the Weighted Aggregator triggers a re-score for affected content items within the next batch cycle.

- **Heuristic override policy:** Heuristics are configured in a versioned Config Service, not hardcoded. Changes to heuristic weights or conditions require a config change review by the Quality Council (see §5.4) and are logged in the heuristic audit trail.

5.4 The Quarterly Quality Council Review

The Quality Council is a cross-functional governance body responsible for reviewing and approving changes to the heuristic system, the penalty table, and the sub-score weighting parameters. It convenes quarterly and produces a written decision record that is stored in the internal engineering wiki and linked from the Config Service changelog.

Membership: Head of Platform Engineering, Head of Trust and Safety, Head of Creator Relations, Head of Editorial, and one rotating external advisor (music industry or academic). Decisions require majority vote; the Head of Trust and Safety holds veto rights over any change to the Authenticity sub-score or penalty table.

6. Promo Detection

Promo Detection is the system responsible for identifying and acting on inauthentic promotional activity — the practice of artificially inflating a content item's engagement signals to improve its Quality Score and grid classification. It operates as an independent ML service with its own data pipeline, model governance process, and human review workflow.

6.1 Threat Model

Inauthentic promotion takes several structurally distinct forms, each with different signal signatures:

- **Velocity bursts:** A large volume of engagement signals (plays, saves, likes) arriving in a compressed time window, inconsistent with the creator's historical trajectory or the content's genre/audience size.

- **Geographic clustering:** Engagement signals concentrated in an anomalously narrow geographic region, particularly if inconsistent with the creator's known fanbase geography or the platform's typical engagement distribution for the genre.
- **Bot fingerprints:** Device characteristics, IP address patterns, user-agent strings, and behavioral timing patterns (e.g., play events with identical duration down to the millisecond) consistent with automated script execution.
- **Coordinated account networks:** Clusters of accounts that exhibit correlated behavior — creating accounts at similar times, following identical patterns of playlist addition, or engaging with exactly the same set of content items across multiple sessions.
- **Cross-platform synchrony:** Engagement spikes that are temporally correlated across multiple platforms simultaneously, suggesting coordinated off-platform promotion through paid engagement services.

6.2 Signal Groups

Promo Detection signals are organized into three groups, each contributing a sub-score to the ensemble classifier:

Group A: Engagement Patterns

- Play velocity (plays per hour, normalized by creator tier)
- Save velocity (saves per hour, normalized)
- Play duration distribution (variance from expected distribution; bot plays cluster at fixed durations)
- Engagement-to-impression ratio anomaly (saves per unique impression; abnormally high ratios are suspicious)
- Re-engagement rate spike (sudden large increase in re-engagement inconsistent with content age)

Group B: Source Attribution

- Geographic concentration (HHI of engagement by country; values above 0.80 are anomalous)
- Platform referral source distribution (proportion of engagement arriving via direct link vs. organic discovery; unusually high direct-link proportion is a signal)
- IP address entropy (Shannon entropy of IP address range diversity; low entropy = few source IPs)
- Known promo service IP ranges (direct match against maintained blocklist; deterministic signal)
- VPN/proxy concentration (proportion of sessions using anonymizing proxies above genre-specific baseline)

Group C: Network and Metadata

- Account age distribution of engaging users (accounts created within 30 days of engagement event; high concentration is suspicious)
- Correlated behavior clustering (graph-based detection of account clusters engaging with identical content sequences)
- Cross-platform timing correlation (temporal cross-correlation with external platform signal feeds, where available via API)
- Creator declaration absence (creator has not submitted a Promotional Campaign Declaration via Connect Hub for content with high-velocity engagement)
- Prior promo detection history (creator or associated accounts have previous confirmed promo violations)

6.3 The Ensemble Classifier

The promo detection model is an ensemble classifier that combines Group A, B, and C sub-scores using a weighted sigmoid function. The architecture is intentionally conservative: a bias term is included to reduce false positives, accepting a slightly

higher false-negative rate in preference to wrongly flagging legitimate organic promotion.

```
# Sub-score computation (each group produces a score in [0, 1]) score_A =
group_A_model.predict(engagement_features) # gradient boosted tree score_B
= group_B_model.predict(source_features) # logistic regression score_C
= group_C_model.predict(network_features) # graph neural network #
Weighted combination raw_score = (0.45 × score_A) + (0.35 × score_B) + (0.20
× score_C) # Weighted sigmoid with bias term (bias shifts decision boundary
right, # reducing false positives at the cost of slightly higher false
negatives) promo_score = sigmoid(raw_score, weight=6.0, bias=-0.8) # Where:
sigmoid(x, w, b) = 1 / (1 + e^(-w×(x + b)))
```

6.4 Decision Thresholds

Promo Score Range	Classification	Automated Action	Notification
0.00 – 0.39	No anomaly	None	None
0.40 – 0.59	Low suspicion	Authenticity sub-score suppressed by 15%; monitoring flag added; increased signal sampling rate for next 7 days	None to creator; internal monitoring alert to Trust and Safety dashboard
0.60 – 0.74	Medium suspicion	Q score penalty –20; COMPACT floor applied regardless of score; 30-day enhanced monitoring window	Creator notification via Connect Hub: "Engagement anomaly detected — review pending"
0.75 – 0.84	High suspicion	Q score penalty –35; content capped at STANDARD regardless of score; human review queue entry created	Creator notification: "Promotional activity review initiated" with appeal link
0.85 – 0.94	Confirmed likely	GHOST class applied immediately; Connect Hub promo flag webhook sent	Formal promo flag notification with evidence summary and appeal rights
0.95 – 1.00	Confirmed	GHOST class applied; promo violation recorded on creator account; human review for escalation decision	Formal promo violation notice with full evidence package; 30-day appeal window opened

6.5 Human Review Workflow

Any content item that reaches the 0.75–1.00 threshold enters the human review queue. The workflow has defined SLAs at each step:

4. **Auto-hold (≤ 2 minutes):** Content automatically transitions to GHOST or STANDARD-cap state (depending on threshold band). A review case is created in the moderation platform, populated with the evidence package: promo score breakdown, signal group sub-scores, engagement timeline visualization, and a comparison to peer cohort distributions.
5. **Reviewer assigned (≤ 4 business hours):** A trained Trust and Safety reviewer is assigned to the case based on workload balancing. The reviewer receives an evidence package PDF (auto-generated from the case data) and access to the raw signal explorer tool.
6. **Decision (≤ 48 business hours from assignment):** Three decision paths are available:
 - **Confirm violation:** GHOST or penalty maintained; promo violation recorded on creator account; creator notified with full evidence.
 - **Clear — organic:** Content restored to organic classification; score recalculated without promo penalties; creator notified of clearance; monitoring flag remains active for 30 days.
 - **Clear — declared promotion:** Creator had submitted or now submits a valid Promotional Campaign Declaration; content restored with "Sponsored" badge applied to grid card.

6.6 Creator Appeal Rights

Creators who believe a promo flag is erroneous have a formal appeal right. Key parameters of the appeal process:

- **Appeal window:** 30 days from the date of the promo flag notification.

- **Appeal submission:** Via the Connect Hub 4-step Appeal Flow (§10.7). Creators may submit evidence of organic promotion activity: campaign receipts, social media posts, press coverage links, distribution records.
- **Appeal review SLA:** Appeals are reviewed by a senior Trust and Safety reviewer within 5 business days.
- **Appeal outcome:** Full reversal (all penalties removed; creator reputation heuristic restored), partial reversal (some penalties reduced; monitoring maintained), or upheld (original decision stands; creator informed of reason).
- **Escalation path:** If a creator believes their appeal was wrongly denied, a second-level review by the Head of Trust and Safety is available within 14 days of the first appeal decision. Second-level decisions are final.

6.7 Model Governance

Governance Dimension	Specification
Retraining cadence	Every 30 days, using confirmed review decisions as labeled training data
Precision target	Precision ≥ 0.92 at the 0.85 threshold boundary (measured on held-out labeled test set)
Recall target	Recall ≥ 0.75 at the 0.60 threshold boundary (acceptable to miss some medium-suspicion cases; prioritize precision at high threshold)
Fairness audits	Quarterly fairness audit measuring false positive rates across creator genre, geographic origin, and account age cohorts; documented disparity above 15% triggers model adjustment
Model versioning	All model versions are stored in the ML model registry with full training metadata; rollback to previous version available within 15 minutes via feature flag
Threshold review	Decision thresholds reviewed at each Quality Council meeting (§5.4) using precision-recall curve analysis

7. Search Strategy

Vibe Atlas's search system is a four-stage pipeline: Query Understanding → Retrieval → Ranking → Result Shaping. The pipeline is designed to handle five distinct search modes, each with different intent profiles, and to integrate Quality Score signals as a first-class ranking factor alongside relevance. Search is not a separate system from the grid — it shares the same scoring infrastructure and produces results on the same quality-weighted basis.

7.1 The Five Search Modes

Mode	Trigger	Example Query	Primary Signal
Keyword	User types a literal query (artist name, track title, album)	"Beyoncé Lemonade"	BM25+ lexical match on title, artist, album fields
Semantic	User types a descriptive or abstract query without exact entity names	"sad rainy day music"	FAISS nearest-neighbor on 768-dim embedding space
Vibe Query	User selects vibe tags from the Vibe Taxonomy (\$7.6)	[High Energy] + [Late Night] + [Euphoric]	Vibe taxonomy vector match + semantic embedding
Similarity	User triggers "more like this" from a content item	(from a playing track card)	FAISS embedding similarity from seed item vector
Contextual	System-initiated; uses session context (time of day, recent plays, location)	(no explicit query)	Session embedding + collaborative filtering + Quality Score

7.2 Stage 1: Query Understanding Service (QUS)

The Query Understanding Service is a Python gRPC microservice that transforms a raw user query string into a structured intent object consumed by the retrieval stage. It runs on every search request and must complete within 30ms to preserve the total P99 latency budget.

QUS Processing Pipeline:

7. **Typo Correction:** Edit-distance-based candidate generation against a phonetic vocabulary index. Corrections are applied silently if confidence ≥ 0.90 ; otherwise the corrected query is surfaced as a "Did you mean?" suggestion.
8. **Intent Classification:** A fine-tuned BERT classifier assigns the query to one of: Entity Lookup, Mood/Vibe Discovery, Genre Exploration, Chart/Trending, or Exploratory. Intent class determines which retrieval path is weighted most heavily.
9. **Entity Extraction:** Named entity recognition identifies artist names, track titles, album names, genre terms, and location references. Recognized entities are linked to the content knowledge graph to resolve ambiguity (e.g., "drake" $\rightarrow$ artist entity, not the name).
10. **Mood and Vibe Tagging:** Adjective and descriptor terms in the query are mapped to Vibe Taxonomy nodes (§7.6) using a fine-tuned classification model. Multi-label output; up to 5 vibe tags assigned per query.
11. **Query Expansion:** Synonym expansion and related-term injection using a music-domain-specific word embedding model. Expansion terms are added to the retrieval query at reduced weight (0.3 $\times$ of primary term weight).
12. **Embedding Generation:** The full query (original + expanded terms) is encoded by the same 768-dim BERT model used to embed content items, producing a query vector for FAISS retrieval.

```
# QUS output schema QueryIntent {      original_query:      string
corrected_query:      string | None    intent_class:      enum [entity_lookup,
mood_vibe, genre, chart, exploratory]   entities: [        { type:
"artist"|"track"|"album"|"genre"|"location",      value: string,
entity_id: string | None }                    ]      vibe_tags:      [string] # taxonomy
node IDs, max 5      query_vector:      float[768]      expanded_terms:
[ { term: string, weight: float } ]          confidence:      float # overall QUS
confidence [0,1] }
```

7.3 Stage 2: Retrieval

The retrieval stage runs two parallel paths — lexical and semantic — and merges their results using Reciprocal Rank Fusion (RRF) before passing the merged candidate set to the ranker.

Lexical Path: BM25+ on Elasticsearch

Elasticsearch is queried with a multi-field BM25+ query across indexed fields: `title` (boost 3.0), `artist_name` (boost 2.5), `album_name` (boost 1.5), `genre_tags` (boost 1.2), `description` (boost 1.0). Entity-linked queries use a term filter on `entity_id` for exact entity matching at maximum precision. The lexical path returns the top 200 candidates.

Semantic Path: FAISS + 768-dim BERT Embeddings

Content items are indexed at publish time with a 768-dimensional BERT embedding computed from a concatenation of title, artist name, genre tags, vibe tags, and description text. The query vector from QUS is used to perform an IVF-PQ approximate nearest-neighbor search in FAISS, returning the top 200 semantic candidates. For Similarity mode queries, the seed item's stored embedding vector is used directly as the query vector.

Reciprocal Rank Fusion Merge

```
# RRF score computation for each candidate item # k = 60 (standard RRF
smoothing constant) # Positions from lexical ranking: rank_lex(item) #
Positions from semantic ranking: rank_sem(item) rrf_score(item) = (1 / (k +
rank_lex(item))) + (1 / (k + rank_sem(item))) # Items appearing in only one
path receive the other path's contribution as 0 # Merged candidate set: top
100 items by rrf_score, passed to ranker
```

7.4 Stage 3: Ranking

The ranking stage applies a LambdaRank model to the merged candidate set of 100 items, producing a final ranked list of up to 50 results returned to the UX layer. LambdaRank is a pairwise learning-to-rank algorithm trained on click-through data with NDCG@10 as the optimization objective.

Feature Groups

Feature Group	Weight	Key Features
Relevance	40%	RRF score from retrieval stage; entity match flag; vibe tag overlap count; query-title edit distance
Quality	30%	Composite Quality Score Q; Authenticity sub-score;

Feature Group	Weight	Key Features
		score band (ordinal encoded); score delta in trailing 24h
Personalization	20%	Collaborative filtering score (user × item affinity from matrix factorization); session context vector similarity; user genre preference alignment; recency of user's interaction with same artist
Freshness	10%	Freshness sub-score F; resurgence boost flag; days since publish (log-normalized)

7.5 Stage 4: Result Shaping

Result shaping applies post-ranking transformations to the ranked list before serialization. These transformations enforce diversity, compliance, and business rules:

- **Diversity enforcement:** No more than 3 content items by the same artist in the top 10 results. Duplicate items are demoted to positions 11+ regardless of rank score.
- **Promo suppression:** GHOST-class items are excluded from results entirely. Items with active promo flags below GHOST threshold are moved to positions 20+.
- **Sponsored slot rules:** Sponsored results (editorially pinned with `sponsored=true`) are capped at 1 item per 10 results and must be labeled with a "Sponsored" badge. They cannot occupy position 1.
- **A/B experiment slots:** The search ranker reserves positions 5, 10, and 15 as configurable experiment slots. These positions receive their content from the A/B test configuration service rather than from the LambdaRank output, allowing controlled experiments on ranking features without affecting the remainder of the result list.

7.6 The Vibe Taxonomy

The Vibe Taxonomy is a 120-node hierarchical ontology that provides a shared vocabulary for describing the emotional, energetic, and contextual character of music and cultural content. It is the backbone of Vibe Query mode and the Vibe Filter Panel UX component.

The taxonomy is organized across three primary dimensions, each with subordinate nodes:

Dimension	Node Count	Example Nodes	UX Color Code
Energy Level	40 nodes	Anthemic, High-Octane, Steady Pulse, Drifting, Still	Orange (#e65c00 spectrum)
Mood	50 nodes	Euphoric, Bittersweet, Melancholic, Triumphant, Tense, Playful, Raw	Blue (#2b5797 spectrum)
Setting	30 nodes	Late Night, Morning Ritual, Festival, Commute, Focus, Intimate, Outdoor	Green (#006633 spectrum)

Content items receive vibe tags at index time via an automated tagging model (fine-tuned audio classifier + text classifier ensemble), supplemented by creator-submitted tags via Connect Hub metadata updates. Human editorial teams may add or override up to 3 taxonomy tags per content item. Tag changes are versioned and logged.

7.7 Autocomplete

Autocomplete suggestions are served from a pre-computed trie data structure stored in Redis. The trie is updated every 30 minutes via a batch job that computes the top 10,000 queries by volume in the trailing 7-day window, supplemented by entity name entries for all verified artists and top-tier content items. Vibe taxonomy node names are also included as autocomplete candidates.

Each autocomplete request returns up to 8 suggestions, categorized as:

- 4 content item matches (track / album titles)

- 2 artist name matches
- 2 vibe tag matches (taxonomy node names)

The P99 autocomplete latency target is **$\leq 80\text{ms}$** end-to-end, measured from request receipt at the API gateway to response delivery. The trie lookup itself must complete in $\leq 5\text{ms}$; the budget for network and serialization is 75ms.

7.8 Search Telemetry and Evaluation Metrics

Metric	Definition	Target	Measurement Cadence
NDCG@10	Normalized Discounted Cumulative Gain at rank 10; primary offline ranking quality metric	≥ 0.72 on labeled evaluation set	After each model retraining
MRR	Mean Reciprocal Rank of first relevant result	≥ 0.58	After each model retraining
Click-Through Rate (CTR)	Proportion of search sessions resulting in at least one content item interaction	$\geq 42\%$ (Phase 2 target)	Daily; 7-day rolling average
Zero-result rate	Proportion of queries returning zero results	$\leq 3\%$	Real-time monitoring
Query abandonment rate	Proportion of searches with no interaction within 60 seconds	$\leq 25\%$	Daily; 7-day rolling average
Autocomplete adoption rate	Proportion of searches where user selected an autocomplete suggestion	$\geq 55\%$	Daily

8. Height Emphasis System

The Height Emphasis System is a dynamic vertical scaling mechanism that modulates the actual rendered height of grid cards beyond their base class multiplier. While grid class determines the nominal footprint of a content item on the grid, Height Emphasis makes the grid feel alive and editorial by allowing cards within the same class to have meaningfully different vertical presences based on real-time signals.

8.1 Purpose and Rationale

A grid in which all STANDARD cards are identical in size is visually static. It communicates only the binary fact of class assignment, not the richness of the signal underneath. Height Emphasis translates the momentum, editorial importance, media richness, and personalization relevance of a content item into a continuous visual dimension — the vertical axis — creating a surface that users can read intuitively without knowing the underlying mechanics.

Design Decision: Continuous vs. Discrete Scaling

An earlier design used discrete height tiers (small/medium/large within each class) rather than a continuous HeightEmphasisFactor. Discrete tiers were rejected because they produced perceptible "jumps" in card height during re-classification events, and because the tier boundaries introduced additional classification parameters that were difficult to govern. Continuous scaling within defined clip bounds [0.80, 1.50] produces smoother transitions and is governed by a single formula rather than a lookup table.

8.2 The Base Height Formula

```
RenderedHeight = BaseRowHeight × ClassMultiplier  
× HeightEmphasisFactor × ResponsiveScaleFactor Where:  
BaseRowHeight = 120px (the fundamental grid row unit)  
ClassMultiplier = { HERO: 4.0, FEATURE: 2.0, STANDARD: 1.0,  
COMPACT: 0.5, GHOST: 0.0 } HeightEmphasisFactor = float in [0.80, 1.50]  
(computed below) ResponsiveScaleFactor = { xs: 0.75, sm: 0.85, md: 0.95,  
lg: 1.00, xl: 1.10 }
```

8.3 HeightEmphasisFactor: The Four Inputs

```
HeightEmphasisFactor = clip(      (0.40 × MomentumFactor)    + (0.30 ×  
EditorialEmphasisScore)      + (0.20 × MediaRichnessFactor)  + (0.10 ×  
PersonalizationFactor),      min=0.80, max=1.50 )
```

Input Factor	Weight	Range	Definition	Update Source
Momentum Factor	40%	[0, 1]	Normalized rate of change of the Quality Score Q over the trailing 24 hours, scaled to [0,1]. A score gaining 10+ points in 24h receives MomentumFact or ≈ 1.0 .	Real-time; recomputed on each score update event
Editorial Emphasis Score	30%	[0, 1]	Curator-assigned emphasis value [0, 1] set at editorial pin time, or 0.5 if no editorial pin is active (neutral default). Value of 1.0 represents maximum editorial amplification.	Set at editorial override creation; persists for override duration
Media Richness Factor	20%	[0, 1]	Composite of: animated cover art (0.8), high-res static artwork $\geq 3000 \times 3000$ (0.6), standard artwork (0.4), text-only card (0.2). Content with multi-media assets (video preview available) scores 1.0.	Computed at content index time; updated on metadata change
Personalization Factor	10%	[0, 1]	User-specific affinity score for this content	Computed per-request by the personalization

Input Factor	Weight	Range	Definition	Update Source
			item, derived from collaborative filtering and session context. 0.0 = no affinity signal; 1.0 = very strong predicted affinity.	service

8.4 Aspect Ratio Policies

Artwork rendering within cards must respect content-type-appropriate aspect ratio policies. Height Emphasis scales the card container; the artwork within adapts according to these rules:

Content Type	Aspect Ratio Policy	Overflow Behavior
Album art (square)	Constrained to 1:1; card height increase creates letterbox above/below with gradient overlay	Background blur of artwork fills letterbox areas
Landscape artwork (16:9)	Width-constrained; height follows natural aspect ratio; excess card height filled with gradient	Bottom gradient into card footer metadata area
Portrait artwork (2:3 or 3:4)	Center-cropped at card width; vertical overflow clipped at card boundary	Hard clip; no overflow permitted outside card bounds
Text card (no artwork)	Full card height used for typography layout; no aspect ratio constraint	Typography scales with HeightEmphasisFactor
Animated cover (GIF / video loop)	Treated as square artwork; animation plays at native frame rate, paused when card is off-screen	Same as album art letterbox policy

8.5 Animation and Transition Rules

Transition Type	Duration	Easing Function	Notes
Initial page load	280ms	ease-out	Cards fade in and scale up simultaneously; staggered by column index (20ms per column)
Re-classification event	450ms	cubic-bezier(0.4, 0, 0.2, 1)	Height change is animated; neighboring cards shift to accommodate; animation deferred until card is in viewport
Editorial override applied	350ms	ease-in-out	Subtle brightness pulse (1.05x) accompanies height change to signal editorial action to user
Momentum spike	600ms	spring (damping 0.7)	Overshoot allowed; card briefly scales to 1.55x before settling at computed value; creates organic feel
Score decay / height reduction	800ms	ease-in	Slower reduction than increase; gradual visual contraction is less disruptive to reading the grid

Reduced Motion Support

All animations must respect the `prefers-reduced-motion` media query. When reduced motion is preferred, all transitions collapse to instantaneous opacity changes (no movement, no scaling). This affects height transitions, card state changes, and the momentum pulse animation. See §10.8 for full accessibility specifications.

8.6 Grid Reflow Strategy

Height changes in one card may necessitate reflow of neighboring cards. The reflow strategy balances visual stability against responsiveness to score updates:

- **Deferred reflow on scroll intersection:** If a card whose height has changed is not currently in the viewport, the height update is applied immediately to the data model but the DOM reflow is deferred until the card enters the viewport intersection observer's threshold (10% visibility). This prevents off-screen layout thrashing from impacting scrolling performance.
 - **Immediate reflow for small deltas:** If the height change is ≤ 1 row unit (120px), the reflow is applied immediately regardless of viewport position. Small adjustments have negligible impact on layout stability.
 - **Reflow batching:** Multiple height updates arriving within a 500ms window are batched into a single reflow pass to prevent multiple consecutive layout recalculations.
 - **Scroll position preservation:** If a reflow event would shift the user's current scroll position, the scroll position is adjusted by the same pixel delta to keep the currently centered content in view. This is critical for maintaining a non-disruptive reading experience during live grid updates.
-

9. Connect Hub Integration

9.1 What is Connect Hub

Connect Hub is the creator-facing portal through which artists, labels, and distributors interact with the Vibe Atlas platform. It provides three primary capabilities: transparency (creators can see how their content is scoring and why), control (creators can submit metadata updates, declare promotional campaigns, and manage their catalog), and recourse (creators can appeal promo flags and moderation decisions). Connect Hub is a separate frontend application that communicates with the Atlas backend through a dedicated API bridge.

9.2 The Bidirectional Bridge

The Atlas ↔ Connect Hub integration is bidirectional. Each direction carries distinct data types with different latency requirements:

Direction	Data Types	Mechanism	SLA
Atlas → Connect Hub	Quality score updates, promo flag alerts, moderation decisions, appeal status changes	HMAC-SHA256 signed webhooks to creator-registered endpoints; fallback to Connect Hub inbox if no endpoint registered	P95 delivery ≤ 5 seconds from trigger event
Connect Hub → Atlas	Creator metadata updates, promotional campaign declarations, appeal submissions, catalog management actions	REST API (PATCH/POST endpoints on Atlas API v1); synchronous acknowledgment; async processing	Metadata reindex P95 ≤ 5 minutes; score update after appeal decision P95 ≤ 5 minutes

9.3 Data Flows and Payload Schemas

Flow 1: Quality Score Update Webhook (Atlas → Connect Hub)

Triggered when a creator's content item Quality Score delta meets or exceeds the creator's configured notification threshold (default: ±10 points; configurable per creator tier).

```
POST {creator_webhook_url} Headers: X-Vibe-Signature: HMAC-SHA256({secret},{payload_body}) X-Vibe-Event: quality_score.updated Content-Type: application/json Payload: {  "event_type": "quality_score.updated",  "event_id": "evt_01HXYZ1234",  "timestamp": "2026-07-09T19:10:00Z",  "content": {    "content_id": "ctr_abc123",    "title": "string",    "artist_id": "art_xyz789"  },  "quality_score": {    "current": 72.4,    "previous": 61.1,    "delta": 11.3,    "band": "gold",    "sub_scores": {      "engagement": 78.2,      "retention": 65.5,      "authenticity": 80.0,      "freshness": 55.3    },    "heuristic_adjustment": 5.0,    "penalties_applied": [],    "score_updated_at": "2026-07-09T19:09:55Z"  },  "grid_class": {    "current": "FEATURE",    "previous": "STANDARD"  } }
```

Flow 2: Promo Flag Webhook (Atlas → Connect Hub)

Triggered when the Promo Detection service assigns a promo score above 0.60 (medium suspicion threshold).

```
POST {creator_webhook_url} Headers: X-Vibe-Signature: HMAC-SHA256({secret},
{payload_body}) X-Vibe-Event: promo_detection.flagged Payload:
{  "event_type": "promo_detection.flagged",  "event_id": "evt_02ABCD5678",
"timestamp": "2026-07-09T19:10:00Z",  "content": {  "content_id":
"ctr_abc123",  "title": "string"  },  "promo_flag": {  "promo_score":
0.87,  "threshold_band": "confirmed_likely",  "action_taken":
"ghost_class_applied",  "score_penalty": null,  "evidence_summary": {
"group_a_score": 0.82,  "group_b_score": 0.91,  "group_c_score":
0.74,  "primary_signals": ["ip_concentration", "play_velocity_burst",
"account_age_cluster"]  },  "review_case_id": "rev_009XYZ",
"appeal_deadline": "2026-08-08T19:10:00Z",  "appeal_url":
"https://connect.vibeatlas.com/appeals/rev_009XYZ"  } }
```

Flow 3: Creator Metadata Update (Connect Hub → Atlas)

```
PATCH /api/v1/content/{content_id}/metadata Authorization: Bearer {jwt_token}
(scope: creator:metadata:write) Request Body: {  "title": "string | null",
"genre_tags": ["string"] | null,  "mood_tags": ["taxonomy_node_id"] | null,
"description": "string | null",  "isrc": "string | null",  "credits": {
"producers": ["string"],  "writers": ["string"],  "features":
["string"]  } | null,  "artwork_url": "string | null",  "release_date":
"ISO8601 date | null",  "declared_promotion": boolean | null } Response: {
"content_id": "ctr_abc123",  "status": "accepted",  "reindex_job_id":
"job_XYZ999",  "estimated_reindex_time": "2026-07-09T19:15:00Z" }
```

Flow 4: Promotional Campaign Declaration (Connect Hub → Atlas)

```
POST /api/v1/creator/{artist_id}/promo-declarations Authorization: Bearer
{jwt_token} (scope: creator:promo:write) Request Body: {  "content_id":
"ctr_abc123",  "campaign": {  "name": "string",  "start_date": "2026-
07-09",  "end_date": "2026-07-23",  "channels": ["paid_social",
"influencer", "playlist_pitching"],  "estimated_spend_usd": 5000,
"territories": ["US", "GB", "CA"],  "declaration_confirmed": true  } }
Response: {  "declaration_id": "decl_001ABC",  "status": "accepted",
"effect": "promo_monitoring_window_extended;
sponsored_badge_exemption_if_within_declared_parameters",
"confirmation_code": "VA-DECL-2026-001ABC" }
```

Flow 5: Creator Appeal Submission (Connect Hub → Atlas)

```
POST /api/v1/appeals Authorization: Bearer {jwt_token} (scope:
creator:appeals:write) Request Body: {  "review_case_id": "rev_009XYZ",
"content_id": "ctr_abc123",  "appeal": {  "statement": "string (max 2000
chars)",  "evidence_urls": ["string"],  // creator-supplied evidence links
"evidence_type": ["organic_campaign_receipt",
"social_media_post",  "press_coverage",
"distribution_report"],  "contact_email": "string",  "agreed_to_terms":
true  } } Response: {  "appeal_id": "appl_777DEF",  "status":
"submitted",  "review_sla": "5 business days",  "case_url":
"https://connect.vibeatlas.com/appeals/appl_777DEF",
"acknowledgment_sent_to": "[email protected]" }
```

9.4 Authentication Model

- **OAuth 2.0 Client Credentials:** Connect Hub backend-to-Atlas API communication uses OAuth 2.0 client credentials flow. Tokens are issued with 1-hour TTL; refresh is handled transparently by the Connect Hub SDK.
- **JWT with Creator/Label Scopes:** All API requests carry a JWT with scopes defined per action. Creator accounts may hold scopes in the `creator:` namespace. Label accounts additionally hold scopes in the `label:` namespace that permit cross-roster actions. Scopes are validated at the API gateway before routing to downstream services.
- **HMAC-SHA256 Webhook Verification:** All outbound webhooks from Atlas to Connect Hub creator endpoints include an `X-Vibe-Signature` header containing the HMAC-SHA256 signature of the request body computed with a per-creator shared secret. Creators are responsible for verifying this signature before processing webhook payloads. The Atlas documentation provides reference implementations in Python, JavaScript, and Go.
- **Webhook retry policy:** Failed webhook deliveries (HTTP 4xx/5xx or timeout after 10 seconds) are retried with exponential backoff: 30s, 2m, 10m, 30m, 2h. After 5 failed attempts, the webhook is marked as failed and the payload is written to the creator's Connect Hub inbox as a fallback notification.

9.5 SLA Commitments

Integration Event	SLA Target	Measurement Point
Quality score update webhook delivery	P95 $\leq$ 5 seconds from score delta event	Time from Postgres trigger $\rightarrow$ webhook HTTP 200 received
Score update webhook after delta	$\leq$ 60 seconds from quality score computation	Time from Q write to Postgres $\rightarrow$ webhook dispatch
Metadata reindex after PATCH	P95 $\leq$ 5 minutes from API acknowledgment	Time from API 202 response $\rightarrow$ Elasticsearch document updated
Promo flag notification	P95 $\leq$ 5 seconds from	Same as quality score

Integration Event	SLA Target	Measurement Point
	promo decision	webhook
Appeal submission acknowledgment	Synchronous; ≤ 3 seconds for API response	Time from POST received $\rightarrow$ 202 response
Appeal decision notification	≤ 5 business days from submission	Operational SLA; monitored via review queue dashboard

10. UX Specification

10.1 Design Principles

The UX of Vibe Atlas is governed by three principles that must be held in balance simultaneously. These principles occasionally tension against each other; where they do, the resolution hierarchy is as listed below.

13. **Discovery feels effortless.** The grid is the primary product. Users should be able to navigate it without reading instructions, understanding scoring, or making explicit choices. The quality and classification system communicates through visual hierarchy, not through labels. A user who has never heard of Quality Scores should be able to accurately intuit that HERO cards are the most important items on the page.
14. **Context is visible but not intrusive.** Users who want to understand why a piece of content is prominently placed should be able to access that information — but it should not interrupt the discovery flow for users who do not. Explainability is accessible via a deliberate interaction (opening the "Why this?" drawer), not surfaced unprompted in the main grid view.
15. **Creators feel respected.** The platform's quality judgment is visible to creators through Connect Hub, not just to users. Creators should never be surprised by a classification decision without a clear, accessible explanation. The UX of Connect Hub creator tools should match the quality of the consumer-facing product.

10.2 Grid Card Anatomy per Class

HERO Card

- **Artwork treatment:** Full-bleed, fills the entire 4×4 card area. Gradient overlay from bottom (opacity 0 → 0.85) to ensure metadata legibility.
- **Metadata visibility:** Artist name (large, white), track title (medium, white), primary genre tag, and top 2 vibe tags visible at rest. Quality band badge ("Platinum") displayed in top-right corner — subtle, not prominent.
- **Momentum indicator:** If $\text{MomentumFactor} \geq 0.80$, a thin animated border (1px, pulsing at 2s interval) appears around the card edge. Color matches the Energy Level vibe tag color. Animation respects `prefers-reduced-motion`.
- **Save / bookmark:** Icon in bottom-right; tap/click toggles saved state; accessible keyboard target (44px minimum).

FEATURE Card

- **Artwork treatment:** Top 60% of card; bottom 40% is metadata area with solid background.
- **Metadata visibility:** Artist name, track title, genre, 1 vibe tag. Quality band badge hidden at FEATURE level (exposed in detail drawer only).
- **Momentum indicator:** Static dot indicator (no animation) if $\text{MomentumFactor} \geq 0.70$.

STANDARD Card

- **Artwork treatment:** Square thumbnail (left-aligned, 48px), text metadata to the right.
- **Metadata visibility:** Artist name and track title only. No vibe tags, no badges at rest.

COMPACT Card

- **Artwork treatment:** Small square thumbnail (32px), single line of text (track title + artist).

- **Metadata visibility:** Truncated single line. No additional metadata at rest.

10.3 HERO Card Special Behaviors

- **Auto-preview on hover:** After 800ms hover dwell time on desktop, a 15-second audio preview begins playing (muted by default; unmute control appears). Preview stops on mouse-out or on explicit close.
- **Expanded metadata drawer:** Clicking the card surface (not the play CTA) opens an inline metadata drawer from the bottom of the card, revealing: full artist name, track title, album (if applicable), release date, genre, all vibe tags, play count (rounded), and the Quality Band indicator with a brief tooltip explaining what the band means. The drawer closes on second click or on Escape key.
- **Momentum pulse animation:** If MomentumFactor ≥ 0.90 ("momentum spike"), the border animation described in §10.2 intensifies to a brighter pulse. The word "Trending" (small caps, subtle) appears in the top-left corner of the card.

10.4 Card States

State	Visual Treatment	Interaction Notes
Default	Full opacity; no overlay	Standard interaction model
Hovered	Brightness +5%; overlay chevron ("play") appears center; save icon revealed	Desktop only; 150ms transition
Focused	2px solid outline (#2b5797); no brightness change	Keyboard navigation; ARIA focus indicator; Tab-navigable
Playing	Animated waveform icon in corner; brightness +10%; play icon replaced with pause icon	Only one card in Playing state at a time across the grid
Saved	Save icon filled / highlighted; no other change to card appearance	Persists across sessions; synced to user account
Sponsored	"Sponsored" label (10pt,	Mandatory per editorial

State	Visual Treatment	Interaction Notes
	gray) in bottom-left; no other elevation change	override with sponsored=true flag; non-removable by client

10.5 Search UX

- **Sticky search bar:** Pinned to the top of the viewport on scroll; 56px height; translucent background at rest, opaque on focus.
- **Typeahead:** Activates after 2 characters; 8 suggestions maximum (4 content items, 2 artist names, 2 vibe tags); each suggestion shows a type badge (Track / Artist / Vibe) and a small thumbnail or icon. Keyboard navigable with ↑/↓ arrow keys; Enter confirms selection; Escape dismisses.
- **Mode switcher:** Horizontal tab bar below the search input (visible after first search): All | Tracks | Artists | Albums | Vibes. Defaults to All. Active tab persists within a session.
- **Voice search:** Microphone icon in search bar; activates device speech-to-text API; transcribed text pre-populates the search field for user confirmation before submission.
- **Empty state:** If search returns zero results, the empty state provides: a "Did you mean?" suggestion (if available from QUS typo correction), 3 editorially curated "Explore instead" suggestions relevant to the query category, and a link to browse the Vibe Taxonomy.

10.6 Vibe Filter Panel

- **Layout:** Bottom sheet on mobile (xs, sm); right-side slide-in panel (296px wide) on md+.
- **Multi-select:** Up to 5 vibe tags may be selected simultaneously. Selecting a 6th tag replaces the oldest selected tag. Selected tags appear as filled chips; unselected as outlined chips.

- **Color-coding:** Chips are color-coded by taxonomy category (Energy Level = orange, Mood = blue, Setting = green). Colors are muted at rest, saturated when selected.
- **Real-time grid update:** Selecting or deselecting a vibe tag triggers a debounced (300ms) grid re-query. The grid transitions with a 200ms fade rather than a hard replace, maintaining positional context.
- **Filter state persistence:** Selected vibe filters are preserved in the URL query string, making filtered views shareable and bookmarkable. On page load, URL-encoded vibe filters are restored and applied immediately.

10.7 Content Detail Page

- **Parallax artwork header:** Full-width artwork image; scrolls at 0.5× page scroll speed, creating a parallax depth effect. On devices respecting `prefers-reduced-motion`, parallax is disabled and the header is static.
- **Play CTA:** Large, primary play button immediately below the parallax header. Initiates streaming. Secondary actions: Save, Share, Add to Playlist.
- **"Why this?" explainability drawer:** Accessed via a subtle link ("Why is this featured?") below the play CTA. Opens an inline panel showing: Quality Band (with band name and definition), a simplified breakdown of the four sub-score contributions (displayed as labeled bars, not exact values), and any active heuristic adjustments (described in plain language, e.g., "This track is boosted because it's a new release from a verified artist" or "This track has high chart recognition"). The drawer never shows exact numeric scores; it shows relative strength (Very High / High / Moderate / Low / Very Low) to remain accessible to non-technical users.
- **Related content:** "More like this" row powered by Similarity search mode (§7.1), surfacing 8 content items by FAISS embedding similarity, filtered through the same promo suppression and diversity rules as standard search.

10.8 Connect Hub Creator Surfaces

Quality Dashboard

- Score gauge: radial gauge showing current composite Q score (0–100) against band boundaries. Current band highlighted.
- Sub-score breakdown bars: four horizontal bars (Engagement, Retention, Authenticity, Freshness) showing normalized sub-score values with percentage labels. Each bar is accompanied by a plain-language explanation of what the sub-score measures.
- 7-day trend chart: line chart of Q score history over the trailing 7 days. Penalty events and heuristic changes are marked as annotations on the chart.
- Catalog overview: tabular list of all creator content items, sortable by Q score, grid class, band, and trend direction.

Score Explainability Panel

- Available for each content item via the catalog detail view.
- Shows: current sub-score breakdown, active heuristic adjustments with plain-language descriptions, active penalties with descriptions and estimated resolution paths, and historical score events (re-score events, flag applications, editorial overrides).
- Designed for non-technical creators; all values shown as relative bands (Very High / High / Moderate / Low / Very Low), not raw numbers. Raw numbers available via an "Advanced view" toggle for technical users.

Promo Flag Alert Center

- Inbox-style interface for all promo flag notifications.
- Each notification shows: content item name, flag date, promo score band, automated action taken, evidence summary (plain language), appeal deadline, and appeal CTA.
- Notification status: New / Under Review / Resolved / Appealed.

4-Step Appeal Flow

16. **Step 1 — Select Content:** Creator selects the flagged content item from their catalog. The review case ID and flag details are pre-populated.
17. **Step 2 — Review Evidence:** Creator reviews the evidence summary provided by the platform (plain-language descriptions of signals detected). Creator may download a full evidence package PDF.
18. **Step 3 — Submit Statement:** Creator provides a written statement (max 2,000 characters) explaining their promotional activity. Creator may attach evidence URLs (campaign receipts, social media posts, press coverage).
19. **Step 4 — Confirm and Submit:** Creator reviews their statement, confirms agreement to the appeal terms, and submits. Confirmation email sent immediately; case status updates available in the Alert Center.

10.9 Accessibility

Requirement	Standard / Specification	Implementation Notes
Color contrast	WCAG 2.1 AA (4.5:1 for body text; 3:1 for large text and UI components)	All card text on gradient overlays must pass contrast check at minimum overlay opacity. Automated contrast testing in CI pipeline.
Keyboard navigation	All interactive elements reachable and operable via keyboard	Grid uses <code>role="feed"</code> with arrow key navigation between cards. Tab order follows reading order. All modal and drawer controls keyboard dismissible.
ARIA roles	WCAG 2.1 AA; ARIA 1.2	Grid: <code>role="feed"</code> . Cards: <code>role="article"</code> with <code>aria-label</code> including artist, title, and grid class. Search: <code>role="combobox"</code> with <code>aria-expanded</code> . Drawers: <code>role="dialog"</code> with <code>aria-modal="true"</code> .
Reduced motion	WCAG 2.1 Success Criterion 2.3.3 (AAA) honored as AA commitment	All animations and transitions disabled when <code>prefers-reduced-motion: reduce</code> is detected. Height

Requirement	Standard / Specification	Implementation Notes
		changes, card transitions, momentum animations, and parallax effects all suppressed.
Touch targets	WCAG 2.5.5 — minimum 44×44px	Save icons, play buttons, tab controls, and filter chips all meet 44px minimum. Cards themselves serve as primary touch targets.
Screen reader support	NVDA + Chrome; VoiceOver + Safari tested	Audio previews announced to screen readers with start/stop state. Playing state changes announced via <code>aria-live="polite"</code> . Grade changes in the grid announced via live region.

11. Implementation Phases

The Vibe Atlas implementation is organized into five phases over a 52-week delivery timeline. Each phase has defined deliverables, go/no-go criteria, and dependency relationships with adjacent phases. Phases are not strictly sequential: Phase 1 begins before Phase 0 is fully complete, but each phase's go/no-go gate must be cleared before the subsequent phase's production deployment.

Phase 0 — Infrastructure (Weeks 1-8)

Objective: Establish the foundational data pipeline and scoring infrastructure

- Kafka cluster provisioning; topic schema design and documentation; Kafka Connect setup for Postgres CDC
- Signal Normalizer microservice: deduplication, schema validation, bot pre-filter
- Quality Score calculator: all four sub-scores; Weighted Aggregator; Redis + Postgres integration
- Elasticsearch cluster setup; initial content index schema; indexing pipeline for existing catalog

- Admin Console v0: score inspection, forced re-score, basic content management
- Atlas API v0: internal-only; health endpoints, score read endpoints, basic content CRUD
- Observability stack: Prometheus, Grafana, OpenTelemetry tracing across all services

Go/No-Go Criteria: Q score computed for 100% of catalog content; P99 score update latency ≤ 60 s on load test; all Kafka topics operational with $\geq 99.5\%$ delivery rate; Elasticsearch search latency P99 ≤ 150 ms.

Phase 1 — Grid MVP (Weeks 9-18)

Objective: Deliver a functional, public-facing discovery grid

- Grid Classifier: classification algorithm (§3.2); async worker pipeline; Redis class state store
- Grid Packing Algorithm: modified bin-packing; anti-clustering rules; column-major layout engine
- Height Emphasis v1: base formula implemented; MomentumFactor and MediaRichnessFactor; EditorialEmphasisScore and PersonalizationFactor deferred to Phase 2
- Promo Detection v1: Group A signals only (engagement patterns); simple threshold-based classifier; manual review queue (no SLA tooling yet)
- Editorial pin system: Admin Console editorial UI; time-boxed override creation; HERO approval gate; audit log
- UX Rendering Layer: grid page assembly; all five viewport breakpoints; card state management; basic card animations
- Responsive grid: all breakpoints tested across device matrix (iOS Safari, Chrome Android, Windows Chrome/Edge, macOS Safari)

Go/No-Go Criteria: Grid packing produces valid layouts with no anti-clustering violations across 10,000 random content sets; P95 re-classification SLA ≤ 90 s; grid

page load P99 $\leq$ 300ms at 50,000 concurrent users (load test); WCAG 2.1 AA compliance confirmed by automated audit (axe-core); Promo Detection Group A precision $\geq$ 0.85 on labeled test set.

Phase 2 — Search and Discovery (Weeks 19-28)

Objective: Deliver full search capability and personalized discovery

- Query Understanding Service: full 6-step pipeline (§7.2); intent classification; entity extraction; vibe tagging; query expansion; embedding generation
- Vibe Taxonomy: 120-node hierarchy designed; content auto-tagging model trained and deployed; creator tag submission via metadata API
- Retrieval: BM25+ Elasticsearch queries (§7.3); FAISS index build and deployment; Reciprocal Rank Fusion merge layer
- LambdaRank model: training data collection begins in Phase 1 via telemetry; model trained and evaluated; four feature groups implemented
- Result Shaping: diversity enforcement; promo suppression integration; sponsored slot rules; A/B experiment slot framework
- Collaborative filtering: matrix factorization model for personalization feature group; user session embedding service
- Autocomplete: trie construction pipeline; 30-minute refresh cadence; 8-suggestion response with category breakdown
- Search telemetry: NDCG@10, MRR, CTR tracking; search evaluation dashboard in Grafana
- Height Emphasis v2: PersonalizationFactor integrated; EditorialEmphasisScore integrated
- Vibe Filter Panel UX: full implementation including debounced real-time grid update and URL state persistence

Go/No-Go Criteria: NDCG@10 $\geq$ 0.72 on labeled evaluation set; autocomplete P99 $\leq$ 80ms; zero-result rate $\leq$ 3%; CTR $\geq$ 38% in beta user cohort; all five search modes operational in production.

Phase 3 — Connect Hub Integration (Weeks 29-38)

Objective: Deliver creator transparency and control tooling

- Connect Hub API bridge: all five data flows with full payload schemas (§9.3); OAuth 2.0 + JWT auth; HMAC-SHA256 webhook signing
- Creator Quality Dashboard: score gauge; sub-score breakdown bars; 7-day trend chart; catalog overview table
- Score Explainability Panel: plain-language sub-score explanations; active heuristic descriptions; historical score events
- Promo Detection v2: Group B and Group C signals integrated; ensemble classifier (§6.3) replacing threshold-based v1; decision thresholds implemented (§6.4)
- Human review queue: SLA tooling; evidence package auto-generation; reviewer assignment system; 48-hour decision SLA enforcement and alerting
- Appeal workflow: 4-step appeal flow UX (§10.8); Appeal submission API; appeal review queue for Trust and Safety team; outcome notification webhooks
- Promotional Campaign Declaration endpoint: creator promo declaration submission; atlas-side declaration processing and sponsored badge logic
- Connect Hub Promo Alert Center: inbox UI; notification status management; evidence package download

Go/No-Go Criteria: Webhook delivery P95 $\leq$ 5s verified under load; metadata reindex P95 $\leq$ 5 minutes; Promo Detection v2 precision $\geq$ 0.92 at 0.85 threshold; appeal submission-to-acknowledgment $\leq$ 3s; human review queue tooling supports $\leq$ 48-hour SLA at projected review volume; first-party security audit of OAuth + HMAC implementation.

Phase 4 — Scale and Optimization (Weeks 39-52)

Objective: Harden at production scale; deliver advanced personalization and platform APIs

- Real-time score updates: sub-60s end-to-end score update pipeline validated at 50,000 concurrent user load; Kafka consumer lag alerting; autoscaling tuned to peak traffic patterns
- Automated model retraining: 30-day retraining pipeline for Promo Detection ensemble; LambdaRank retraining cadence; model registry integration; automated evaluation gates before production promotion
- Grid personalization v2: per-user grid layout ordering (not just Height Emphasis); A/B testing framework for personalization algorithms; opt-out mechanism for personalized discovery
- Atlas Public API v2: versioned public API for third-party integrations; rate limiting; developer documentation; API key management
- Geo-redundancy: multi-region deployment for Kafka, Redis, and Elasticsearch; active-passive failover for Postgres; latency-based routing at API gateway
- WCAG 2.1 AA audit: third-party accessibility audit across all consumer-facing surfaces and Connect Hub; remediation of all identified issues
- Fairness audit: first Promo Detection fairness audit (§6.7); documentation of findings; remediation plan if disparities exceed 15%
- Quality Council: first meeting; heuristic review; penalty table review; sub-score weight review; decision record published

Go/No-Go Criteria: Platform sustains 50,000 concurrent users with P99 API latency $\leq 300\text{ms}$ in 72-hour sustained load test; geo-redundancy failover tested with RTO ≤ 5 minutes; WCAG 2.1 AA third-party audit passed with zero critical findings; automated retraining pipeline executes end-to-end without human intervention; Public API v2 launched with full documentation.

Phase Gate Governance

Each phase gate is a formal go/no-go decision meeting with required attendees: Engineering Lead, Product Lead, QA Lead, and Head of Trust and Safety (for phases

involving moderation systems). The gate decision is documented in the project wiki with:

- Go/no-go criteria status (met / partially met / not met) with evidence links
- Open risks and mitigations accepted by leadership
- Feature flag rollback plan: all Phase N features are deployed behind feature flags and can be rolled back to Phase N-1 state within 15 minutes by toggling flags in the Config Service
- Known issues deferred to the next phase, with explicit acceptance sign-off

Cross-Phase Dependencies

Dependency	From Phase	Required By Phase	Critical Path
Kafka pipeline + Signal Normalizer operational	Phase 0	Phase 1, Phase 2, Phase 3	Yes — all downstream scoring depends on this
Quality Score (all 4 sub-scores) in Redis	Phase 0	Phase 1 (Grid Classifier), Phase 3 (Connect Hub)	Yes
Content embedding index in FAISS	Phase 2	Phase 2 (Similarity search), Phase 4 (personalization v2)	Phase 2 internal
LambdaRank training data (telemetry from Phase 1)	Phase 1	Phase 2 (model training)	Yes — 8+ weeks of telemetry needed before Phase 2 model training
Promo Detection v1 (Group A)	Phase 1	Phase 3 (v2 builds on v1 infrastructure)	Partial
Connect Hub API bridge	Phase 3	Phase 4 (Public API v2 references Connect Hub contracts)	No — Public API v2 can proceed independently
Vibe Taxonomy (120 nodes)	Phase 2	Phase 2 (Vibe Filter Panel), Phase 3 (creator tag submission)	Phase 2 internal; must be complete before Vibe Filter Panel ships

12. Open Questions and Future Work

The following areas represent deliberate decisions to defer architectural design beyond the 52-week delivery scope. Each represents a meaningful extension of the platform's capabilities that requires either significant additional infrastructure investment, resolution of unsettled technical questions, or dependencies on external platform decisions.

12.1 Federated Quality Scoring Across Platform Partners

The current Quality Score Engine is a closed system: all signals originate on the Vibe Atlas platform. A federated scoring model would allow partner platforms — distributors, other streaming services, live ticketing platforms — to contribute verified, privacy-preserving signal contributions to a content item's Quality Score. The primary open question is the trust model: how do we verify the integrity of a signal contribution from an external party without exposing raw user data, and what prevents a partner from injecting artificially elevated signals?

Candidate approaches include differential privacy mechanisms for signal aggregation, zero-knowledge proofs of engagement attestation, or a tiered trust model where partner signals are heavily discounted and subject to the same Promo Detection pipeline as first-party signals. This work requires both a technical proof-of-concept and a commercial/legal framework before architectural commitment.

12.2 Real-Time Collaborative Playlist Integration

Vibe Atlas's current architecture treats playlists as static metadata (content appears in user playlists; playlist inclusion is one input to the Community heuristic). A real-time collaborative playlist model — where multiple users can co-edit a playlist simultaneously, and where collaborative listening sessions are visible to other users — would significantly enrich the social layer of the platform. Key design questions: How does collaborative listening affect session context signals for the personalization model? How do we render live collaborative activity in the grid without disrupting individual discovery flow? How do we handle conflicts in simultaneous playlist edits?

12.3 Generative AI Vibe Matching (Natural Language → Vibe Vector)

The current Vibe Taxonomy is a 120-node fixed hierarchy. A generative AI vibe matching system would allow users to describe what they want in completely freeform natural language ("music that sounds like driving through the desert at dusk"), which would be transformed into a continuous vibe vector in the embedding space rather than a discrete taxonomy node selection. This would replace the current Semantic search mode's crude approximation of mood-based queries with a purpose-built natural language interface.

The open question is evaluation: how do we measure whether a generative vibe vector accurately captures user intent, given the subjective nature of mood and setting descriptions? A human evaluation framework for vibe matching quality would need to be designed before this feature could graduate from research to production.

12.4 Creator-Owned Quality Signal Contributions

An experimental direction in which creators contribute verified signal attestations directly to the Quality Score pipeline — for example, live concert attendance data from a ticketing partner, merchandise sales data from a storefront integration, or verified radio airplay counts from a performing rights organization. The architectural challenge is integrating heterogeneous, creator-controlled signal sources while maintaining the integrity of the scoring system against manipulation via false attestations.

12.5 Multi-Modal Content Support (Video, Podcasts, Live Streams)

Vibe Atlas is currently designed for music and audio-primary content. Extension to video content (music videos, visual albums, live concert films), podcast content (long-form audio without a traditional track structure), and live streams (ephemeral content with fundamentally different engagement patterns) would require significant changes to the scoring model. Completion Rate as a retention signal behaves differently for a 90-minute live stream than for a 3-minute track; the Freshness decay model has no natural equivalent for ephemeral live content; and

the grid card anatomy was designed around square album artwork, not landscape video thumbnails.

Multi-modal support is the most significant future architecture work item and would effectively constitute a new major version of the scoring and classification systems. It is deferred pending a product decision on content type expansion strategy.

Document Status and Review Process

This document is Version 1.0 and represents the architectural design as of the Phase 0 kickoff. It will be updated at each phase gate with decisions made during implementation that diverge from the designs specified here. All changes require review and approval by the Engineering Lead and Product Lead. Significant architectural changes — defined as changes that affect inter-system contracts (API schemas, Kafka topic schemas, scoring weights, or classification algorithm logic) — additionally require sign-off from the Head of Platform Engineering. The document change history is maintained in the engineering wiki alongside this document.